E-commerce User Behaviour Analysis (Cosmetics Dataset)

An analytical project using e-commerce user interaction data

Introduction

With the growth of online shopping, understanding how users interact with e-commerce platforms has become increasingly important. This project analyzes user behavior in an online cosmetics store using event level data. The goal is to understand how users move from browsing products to making a purchase, and to identify where users drop off in the process.

Dataset and Approach

The dataset consists of user interaction events such as product views, cart additions, and purchases. Each row represents an action taken by a user at a specific time.

To make the analysis manageable, a subset of the data was used. The data was cleaned by converting timestamps into a usable format, filtering only relevant events (view, cart, purchase), and removing any inconsistent values.

The analysis focuses on these main areas:

- Event distribution
- Funnel analysis
- Time to purchase
- User segmentation
- Price behavior

Event Distribution

The first step was to understand how users interact with the platform overall. Most of the activity comes from product views, followed by cart additions, and then purchases.This shows that while many users browse products, only a smaller portion move forward to actually buying them. This is expected in most e-commerce platforms, but it also highlights the importance of understanding where users drop off.

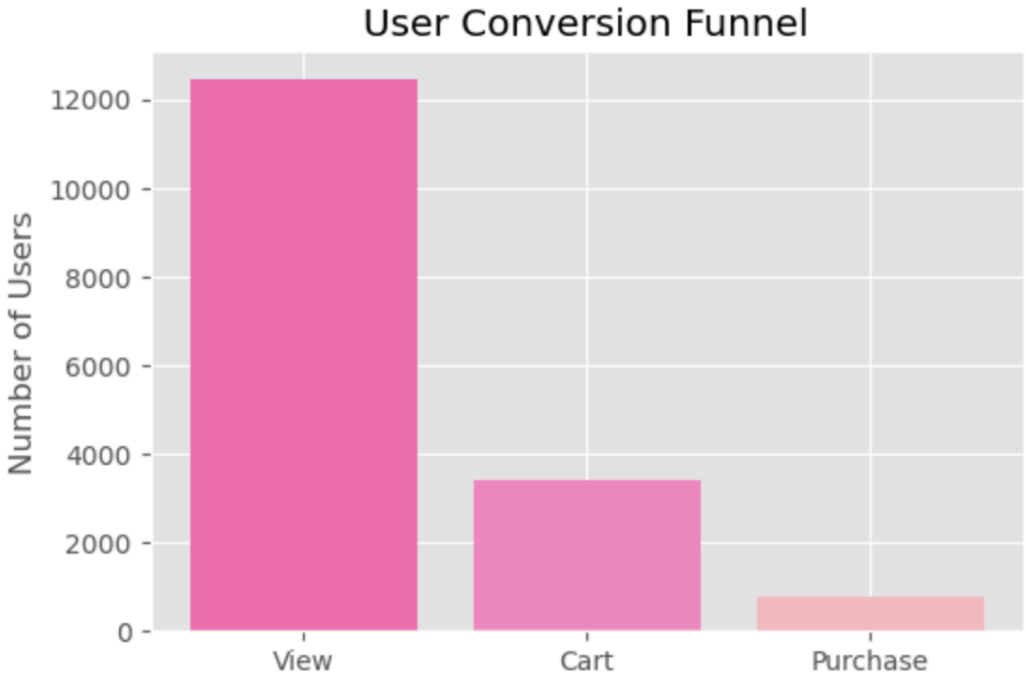
Funnel Analysis

To better understand user behavior, a funnel was created showing how users move from viewing products to adding them to their cart and finally making a purchase.

Around 27% of users who view products go on to add items to their cart, which shows strong initial interest. However, only about 23% of those users complete a purchase, resulting in an overall conversion rate of roughly 6%.

This was interesting because I initially expected more users to drop off earlier in the process, but instead most users reached the cart stage. The main drop off happens later, between the cart and purchase stages.

This suggests that the biggest challenge is not getting users interested, but helping them complete the final step of the purchase.This raises the question of whether friction at the final stage is driven by pricing, decision uncertainty, or lack of trust.



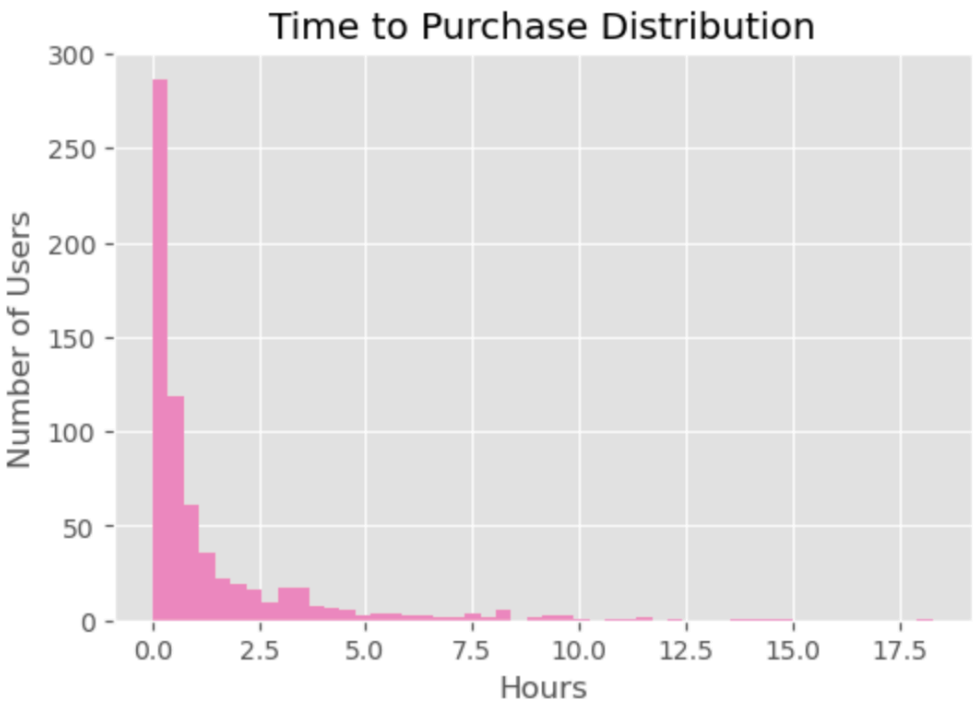
Time to Purchase Analysis

The time between a user's first product view and their purchase was analyzed to understand how quickly decisions are made.

The results show that most users complete their purchase within a short time, often within the first hour. This suggests that many users make quick decisions.

However, some users take longer, sometimes several hours, before completing a purchase. This likely means they are comparing products or returning later before making a decision.

Overall, this shows that there are two types of behavior: some users make quick decisions, while others take more time before buying.



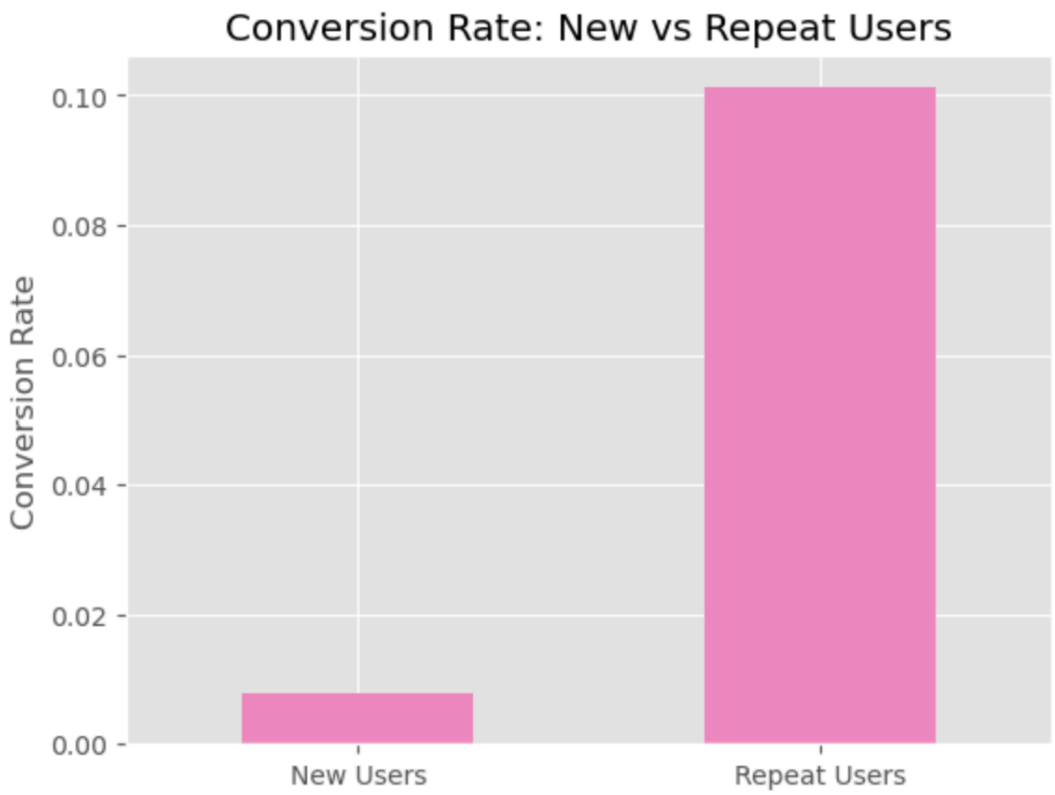
Repeat User Behavior

Users were divided into new users and repeat users based on their activity.

New users mostly browse products, with very few completing purchases. In contrast, repeat users show much higher engagement and are far more likely to add items to their cart and complete purchases.

Repeat users are approximately 13 times more likely to convert than new users, highlighting the strong impact of familiarity and trust on purchase behaviour.

This shows that repeat users are significantly more valuable, as they contribute more to actual conversions. It also suggests that familiarity and trust in the platform play an important role in user behavior.



Price Analysis

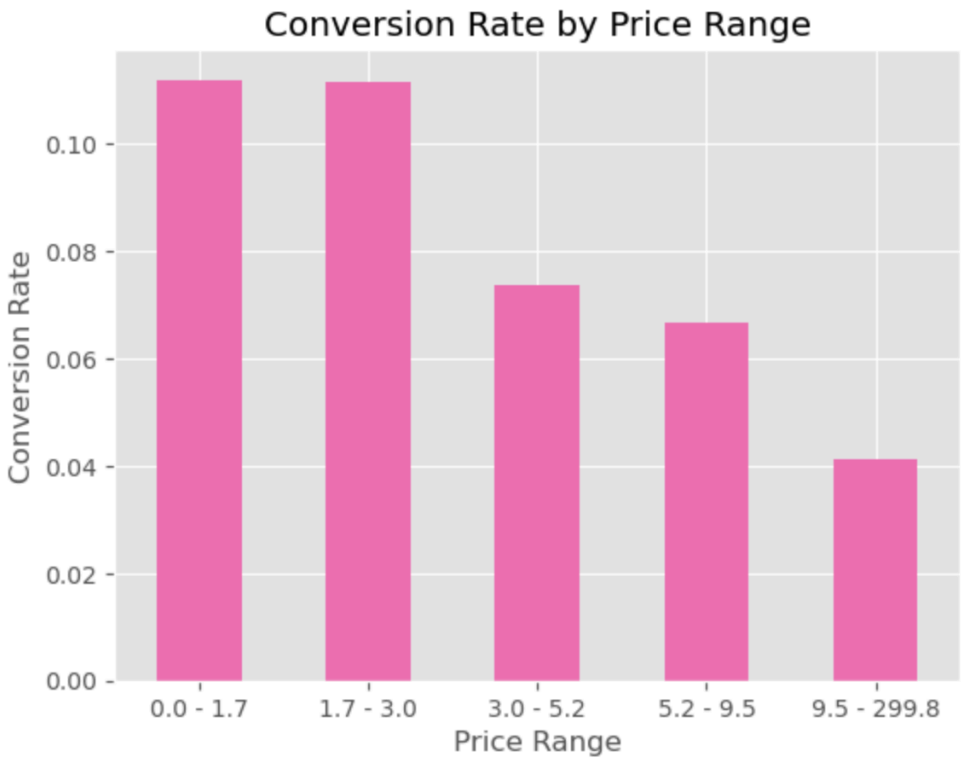
The price analysis shows that users often view higher priced products, but tend to purchase cheaper ones.

A simple logistic regression model further supports this relationship, showing that higher prices are associated with a lower likelihood of conversion.

This suggests that while users are interested in more expensive items, they may hesitate when it comes to actually buying them. Instead, they choose more affordable options.

Conversion rates in higher price ranges are approximately 63% lower than in lower price ranges, indicating a strong negative relationship between price and purchase likelihood.

This also helps explain some of the drop off in the funnel, especially at the final stage, where price may influence the decision to complete a purchase.



Conclusion

This analysis shows that while many users engage with products and progress to the cart stage, only a smaller proportion complete purchases, resulting in an overall conversion rate of approximately 6%. This indicates that the primary drop-off occurs at the final stage of the funnel. Although users demonstrate strong initial interest, a significant number do not follow through, suggesting that improving the checkout or decision stage could significantly increase overall conversion rates.

The time-to-purchase analysis further reveals two distinct behavioural patterns. Many users complete purchases quickly, often within the first hour, indicating fast decision-making. In contrast, a smaller group of users take longer, suggesting more deliberate behaviour such as comparing options or returning later. This highlights variation in user intent and decision processes across the platform.

The price analysis shows that conversion rates decline as price increases, with higher price ranges exhibiting approximately 63% lower conversion compared to lower price ranges. This indicates that price sensitivity plays an important role in influencing final purchase decisions and may contribute to drop-offs at later stages of the funnel.

In addition, repeat users exhibit significantly higher engagement and conversion compared to new users, being approximately 13 times more likely to convert. This highlights the importance of familiarity and trust in driving purchases.

Overall, this analysis demonstrates how user behaviour evolves across different stages of the purchase journey and highlights the measurable impact of factors such as price, decision timing, and repeat interaction on conversion outcomes.

To extend this analysis, I would aim to model the relationship between early user actions and conversion outcomes using predictive methods. In particular, I would examine whether early interaction patterns can be used to predict retention, and quantify the impact of factors such as price and engagement on conversion likelihood.

The full code for this project is available on GitHub: [E-commerce User Behaviour Analysis (Cosmetics Dataset)]